METRIC

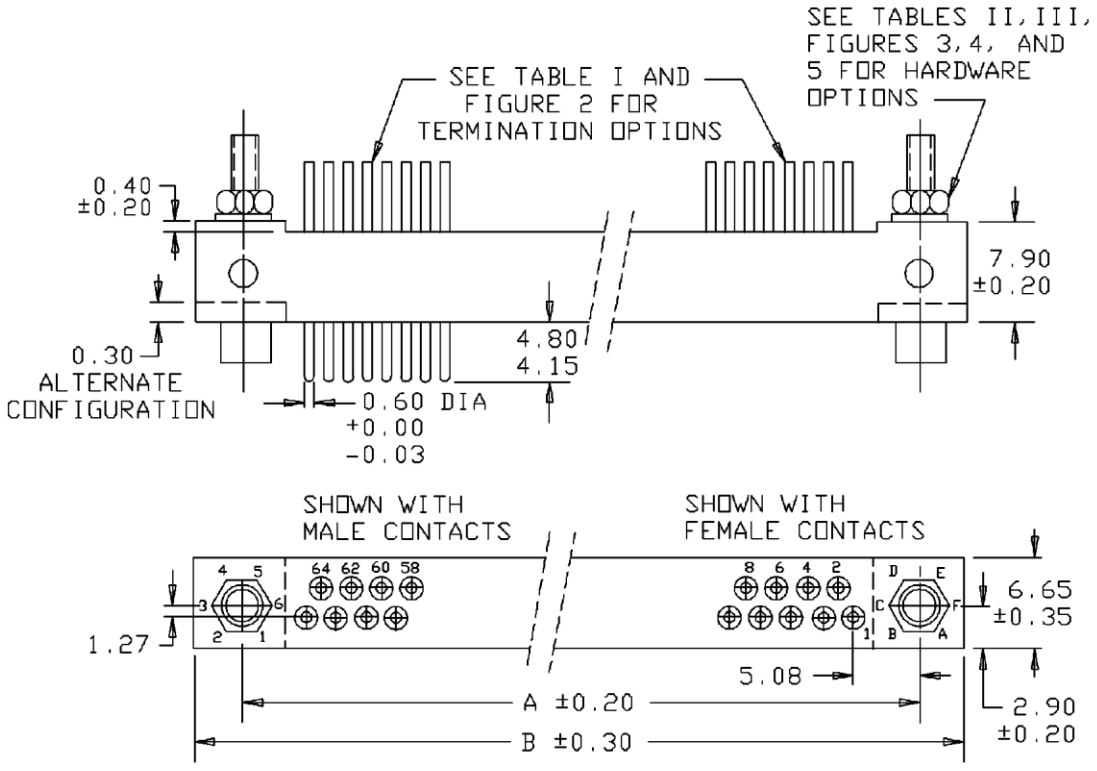
MIL-DTL-55302/159D
w/AMENDMENT 1
20 July 2011
SUPERSEDING
MIL-DTL-55302/159D
8 July 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, RECEPTACLE, 17, 29, 33, 41, 53, AND 65 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS, 2.54 MM X 1.27 MM OFFSET GRID, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.



NOTE: Dimensions are in millimeters.

FIGURE 1. Connector receptacle.

AMSC N/A

FSC 5935

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RECEPTACLE TERMINAL STYLES

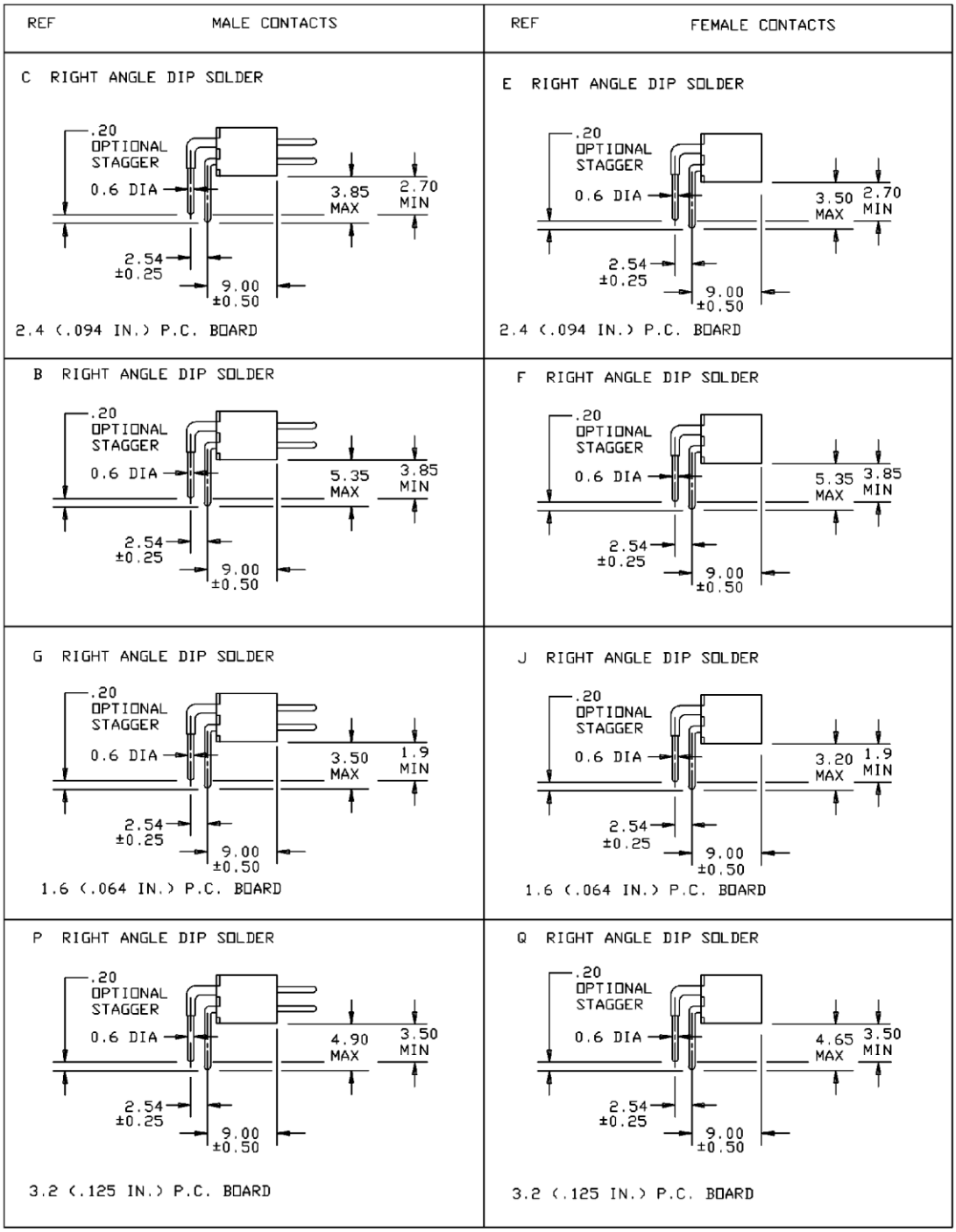


FIGURE 2. Connector termination styles.

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RECEPTACLE TERMINAL STYLES

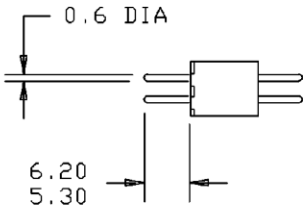
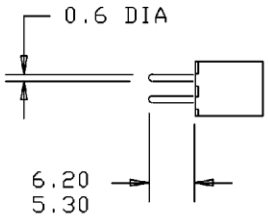
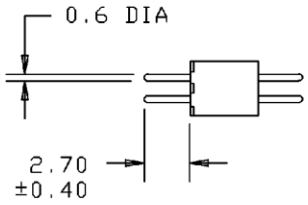
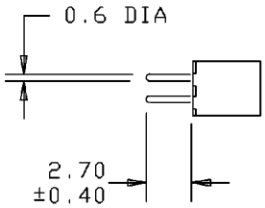
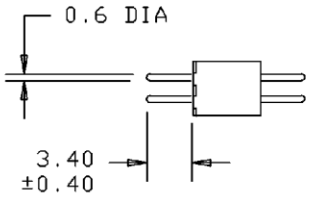
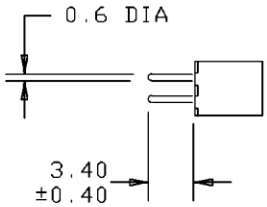
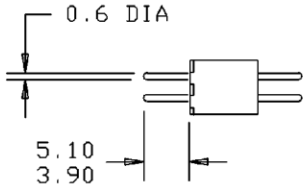
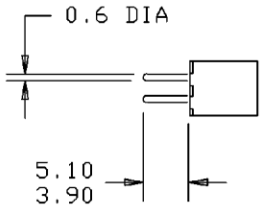
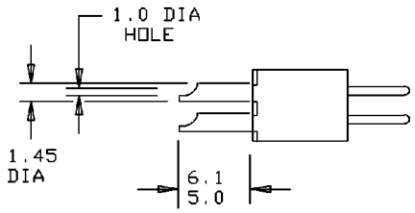
REF MALE CONTACTS	REF FEMALE CONTACTS
A STRAIGHT DIP SOLDER 	D STRAIGHT DIP SOLDER 
T STRAIGHT DIP SOLDER  1.6 (.064 IN.) P.C. BOARD	U STRAIGHT DIP SOLDER  1.6 (.064 IN.) P.C. BOARD
N STRAIGHT DIP SOLDER  2.4 (.094 IN.) P.C. BOARD	Z STRAIGHT DIP SOLDER  2.4 (.094 IN.) P.C. BOARD
V STRAIGHT DIP SOLDER  3.2 (.125 IN.) P.C. BOARD	W STRAIGHT DIP SOLDER  3.2 (.125 IN.) P.C. BOARD

FIGURE 2. Connector termination styles – Continued.

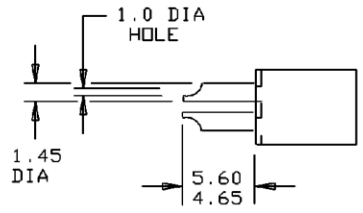
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RECEPTACLE TERMINAL STYLES

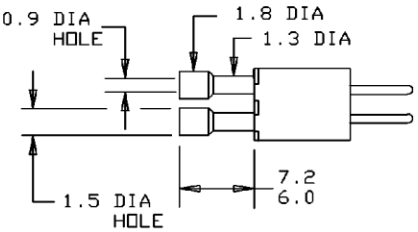
REF	MALE CONTACTS	REF	FEMALE CONTACTS
S	SOLDER CUP	L	SOLDER CUP
M	DOUBLE CRIMP	H	DOUBLE CRIMP
R	CRIMP	K	CRIMP
		Y	WRAP POST



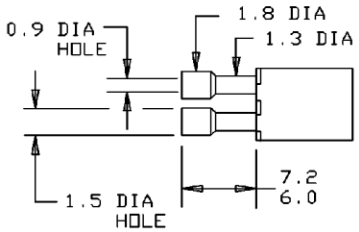
FOR UP TO 22 AWG.



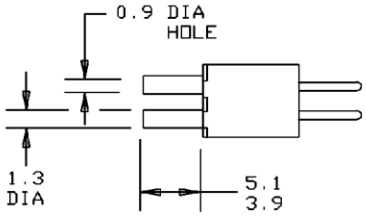
FOR UP TO 22 AWG.



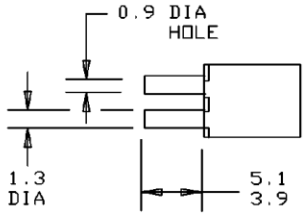
FOR 22,24 AND 26 AWG.



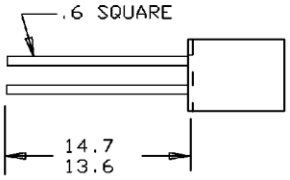
FOR 22,24 AND 26 AWG.



FOR 22,24 AND 26 AWG.



FOR 22,24 AND 26 AWG.

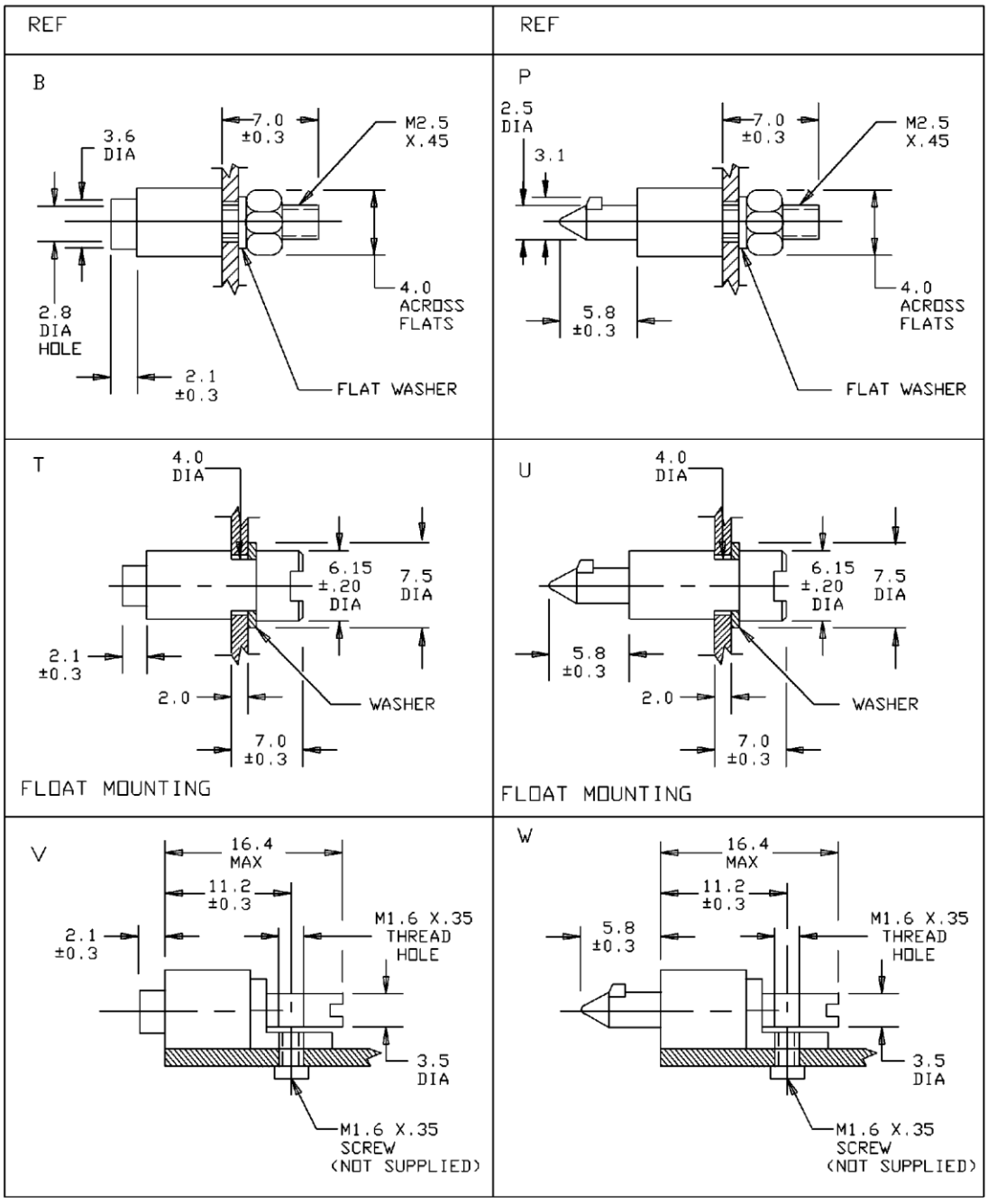


NOTE: Dimensions are in millimeters.

FIGURE 2. Connector termination styles – Continued.

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RECEPTACLE MOUNTING STYLES



NOTE: Dimensions are in millimeters.

FIGURE 3. Mounting styles.

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RECEPTACLE LOCKING HARDWARE

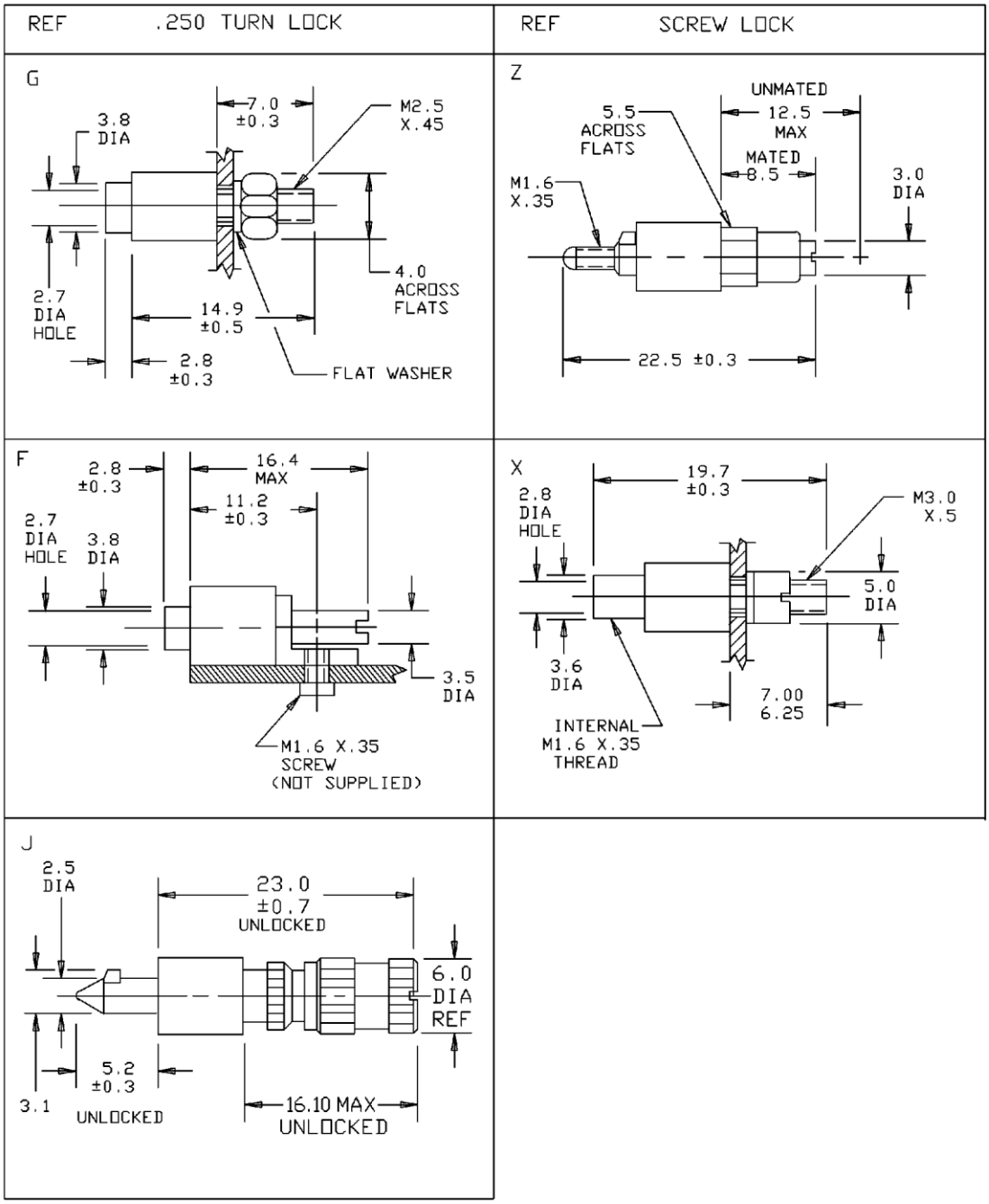


FIGURE 4. Locking hardware.

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NOTES:
1. Dimensions are in millimeters.
2. This is a metric design.
3. Unless otherwise specified, tolerance is ± 0.10 millimeter.
4. These connectors mate with MIL-DTL-55302/162.
5. See table II for mating.
6. Contact identification may be located adjacent to cavity on face or side and may be rotated 180°.
7. Crimp contacts are shipped not inserted into insulator.
8. Use crimp tool M22520/2-01 for R and K terminal styles with Daniels manufacturing positioner K547.
9. Use crimp tool M22520/2-01 for M and H terminal styles with Daniels manufacturing positioner K547 and K640.

FIGURE 4. Locking hardware – Continued.

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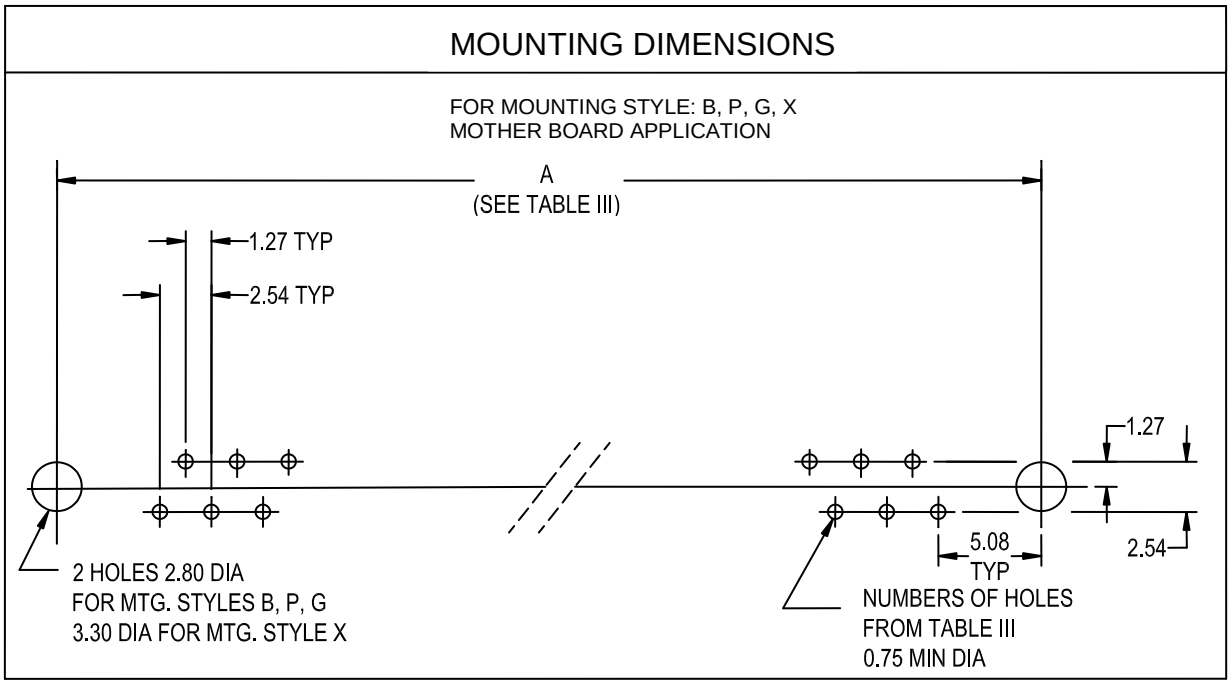


FIGURE 5. Mounting dimensions.

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MOUNTING DIMENSIONS

FOR MOUNTING STYLE : F . V. W
DAUGHTER BOARD APPLICATION

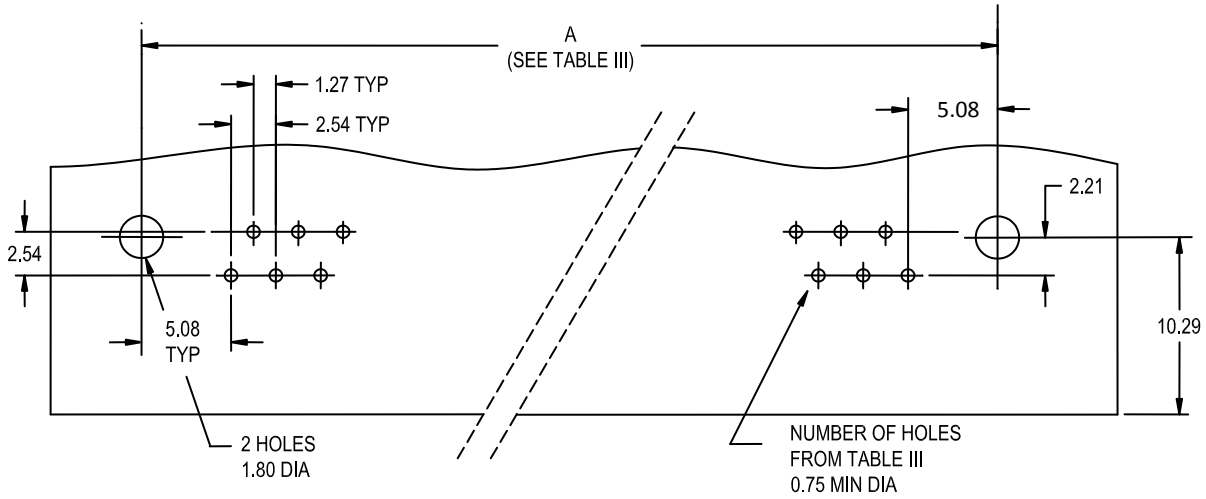
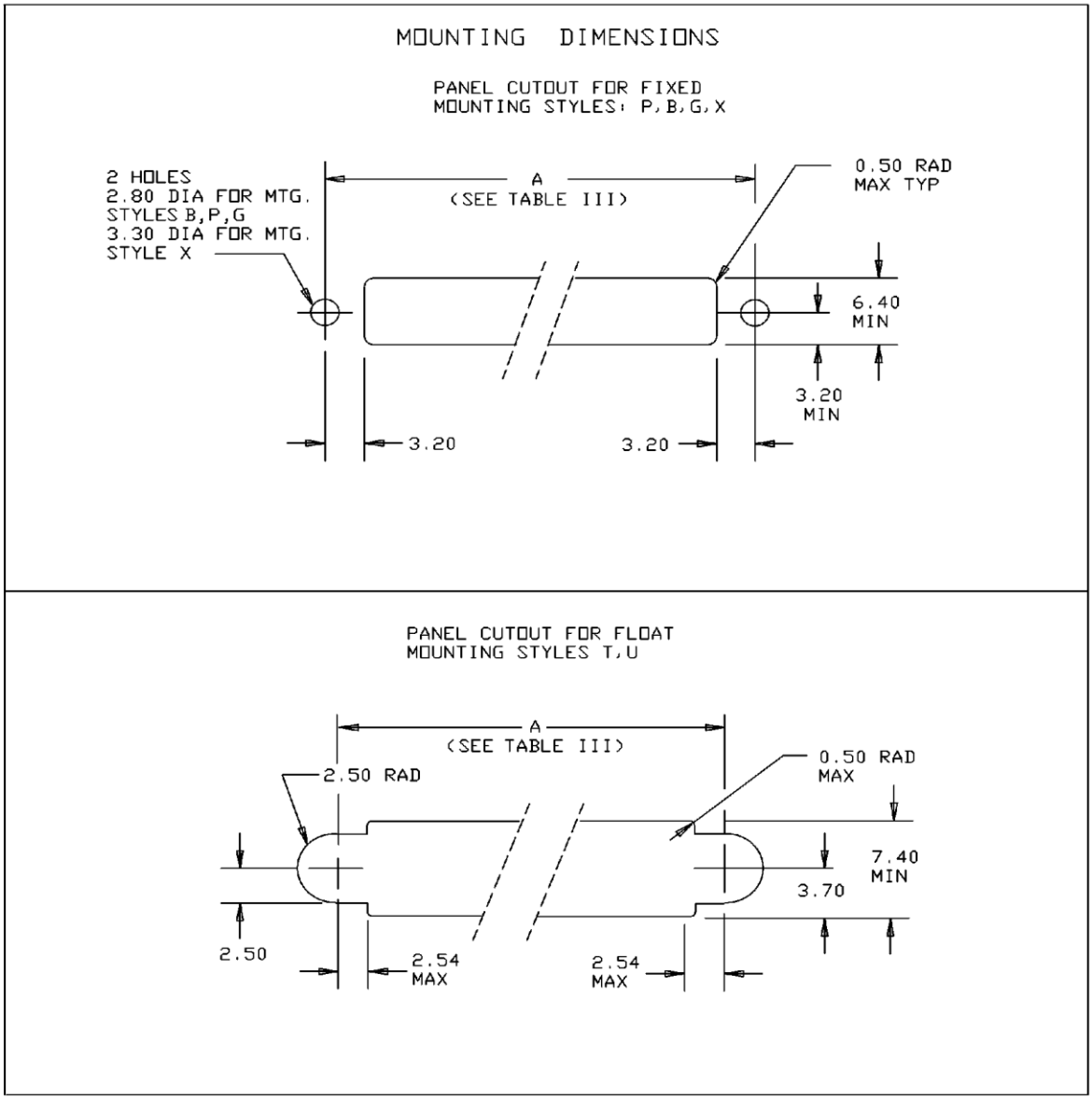


FIGURE 5. Mounting dimensions – Continued.

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NOTES:
1. Dimensions are in millimeters.
2. Tolerances are noncumulative.

FIGURE 5. Mounting dimensions – Continued.

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TABLE I. Number of contacts.

Number of contacts	Dimension (in millimeters)	
	A	B
17	30.50	38.50
29	45.70	53.70
33	50.80	58.80
41	60.90	68.90
53	76.20	84.20
65	91.40	99.40

TABLE II. Mating references.

Type	Will only mate with (see note)	Locking method
B	P,U,W	None
F	J,K	Push 1/4 turn
G	J,K	Push 1/4 turn
J	F,G	Push 1/4 turn
P	B,T,V	None
T	P,U,W	None
U	B,T,V	None
V	P,U,W	None
W	B,T,V	None
X	Z	Screw
Z	X,Y	Screw

NOTE: These connectors mate with MIL-DTL-55302/162.

TABLE III. Mounting references.

Number of contacts	A
17	30.50
29	45.70
33	50.80
41	60.90
53	76.20
65	91.40

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REQUIREMENTS

Design and construction:

Dimensions and configuration: See figures 1 through 5 and tables I, II and III.

Material:

Insulators: Diallyl phthalate in accordance with ASTM D5948.

Guides and hardware: Brass in accordance with ASTM B36/B36M, ASTM B16/B16M, ASTM B124/B124M, ASTM B453/B453M, or ASTM B455; nickel plated in accordance with SAE AMS-QQ-N-290 or AISI 300 series stainless steel or equivalent in accordance with ASTM A582/A582M, passivated in accordance with SAE AMS2700.

Plating: Plating in accordance with MIL-DTL-55302 or as noted.

Mating surface (pin and wire): Gold in accordance with ASTM B488, class 1.27, type II, code C over nickel in accordance with SAE AMS-QQ-N-290, 30-150 microinches.

Terminations: Solderable areas (including crimp barrels and wire wrap terminations) shall be gold in accordance with ASTM B488, type III, code A, class 0.254 minimum thickness or gold in accordance with ASTM B488, type II, code C, class 0.51 over nickel in accordance with SAE AMS-QQ-N-290, 30 to 150 microinches thick. Solderable areas must meet the requirements of MIL-STD-202, method 208.

All other surfaces: Gold in accordance with ASTM B488, type II, code C, class 0.127 minimum over nickel in accordance with SAE AMS-QQ-N-290, 30-150 microinches.

Contact identification: See figure 1.

Contact separation forces:

Minimum force: 0.14 newton (0.50 ounce) with a 0.5800 millimeter (.0228 inch) +0.005, -0.00 diameter pin after being cycled 3 times.

Note: Steel pin test surface roughness to be 0.025-0.25 micron rms.

Unmating: The maximum force of a connector assembly shall not exceed 0.53 newton (1.90 ounces) multiplied by the number of contact positions.

Mating: The maximum force of a connector assembly shall not exceed 0.70 newton (2.5 ounces) multiplied by the number of contact positions.

Contact retention minimum: functional value 22.2 newtons (5 pounds); destructive test 35.3 newtons (8 pounds) maximum design and qualification value.

Contact rating: 5 amperes per line.

Durability: 2,000 cycles minimum.

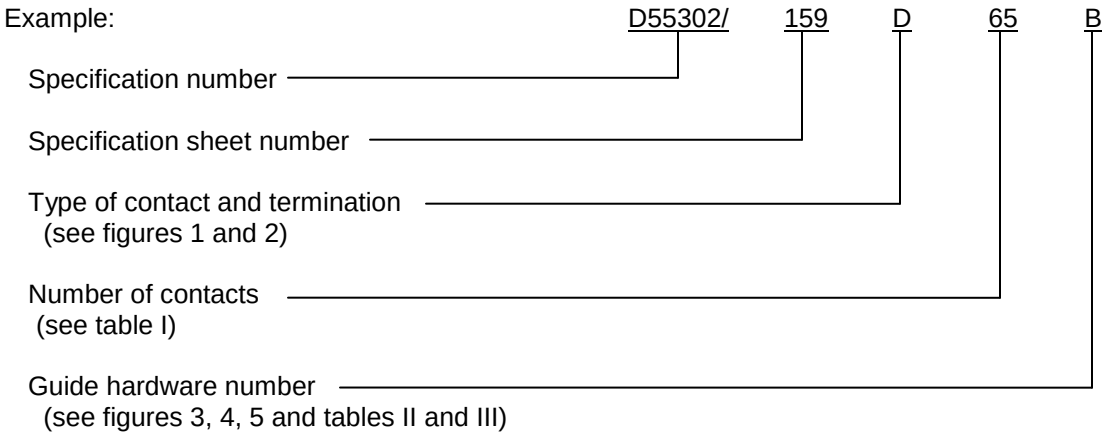
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Dielectric withstanding voltage:

Sea level: 1,400 V ac rms test.
High altitude: 300 V ac rms test.

Keying: Rotation of guides, 6 positions each, 36 possible combinations.

Connector Part or Identifying Number (PIN):



Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/162
MIL-STD-202
ASTM A582/A582M
ASTM B16/B16M
ASTM B36/B36M
ASTM B124/B124M
ASTM B453/B453M
ASTM B455
ASTM B488
ASTM D5948
SAE AMS-QQ-N-290
SAE AMS2700

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CONCLUDING MATERIAL

Custodians:
Army – CR
DLA – CC

Preparing activity:
DLA – CC

(Project 5935-2010-126)

Review activities:
Army – AR, AT, AV, CR4, MI

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